

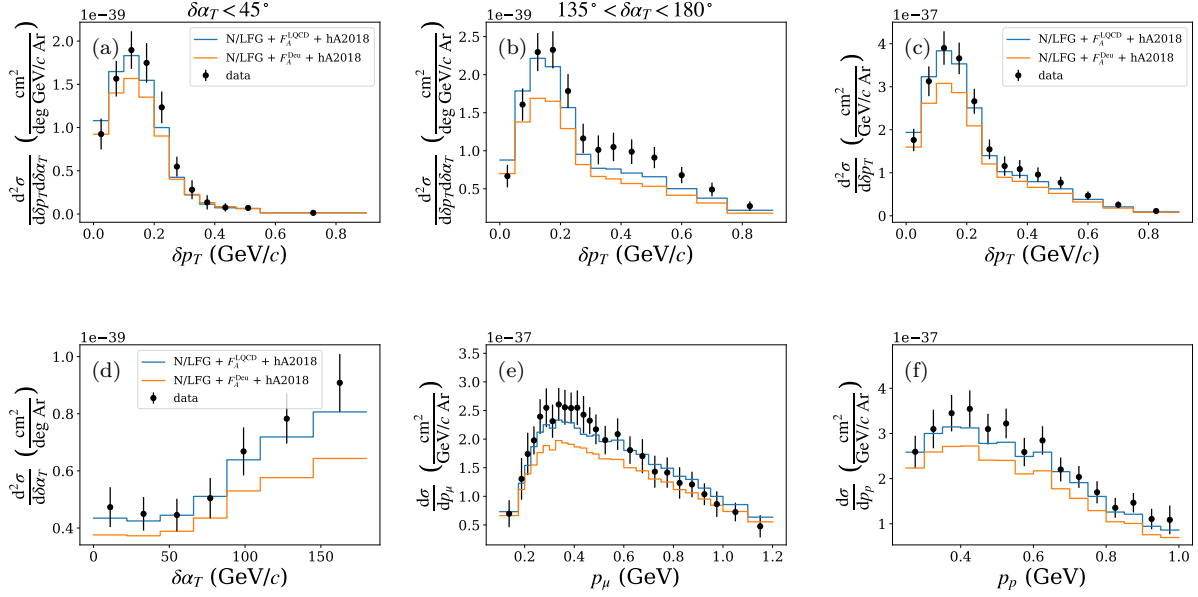


Figure 1: Effect of the nucleon axial form factor on the MicroBooNE CC1 μ 1p / CC1 μ Np argon cross sections with the N/LFG ground state and hA2018 FSI held fixed: F_A^{LQCD} (blue) versus F_A^{Deu} (orange), compared with MicroBooNE data (black points). Panels: (a) δp_T for $\delta\alpha_T < 45^\circ$ and (b) for $135^\circ < \delta\alpha_T < 180^\circ$ (two slices of the 2D δp_T - $\delta\alpha_T$ measurement); (c) δp_T ; (d) $\delta\alpha_T$; (e) p_μ ; (f) p_p . The LQCD form factor raises the predicted cross section and lowers χ^2/nfd in every observable (see Table 1), favouring F_A^{LQCD} .

Table 1: χ^2/nfd (p -value) relative to MicroBooNE data for Fig. 1 (N/LFG ground state and hA2018 FSI held fixed).

Configuration	δp_T vs. $\delta\alpha_T$	δp_T	$\delta\alpha_T$	p_μ	p_p
N/LFG + F_A^{LQCD} + hA2018	35.98/49 (0.92)	6.78/13 (0.91)	3.47/7 (0.84)	16.29/26 (0.93)	16.05/15 (0.38)
N/LFG + F_A^{Deu} + hA2018	44.21/49 (0.67)	9.59/13 (0.73)	10.19/7 (0.18)	20.89/26 (0.75)	19.95/15 (0.17)

Table 2: χ^2/nfd (p -value) relative to MicroBooNE data for Fig. 2 (spectral-function ground state and INCL FSI held fixed).

Configuration	δp_T vs. $\delta\alpha_T$	δp_T	$\delta\alpha_T$	p_μ	p_p
SF + F_A^{LQCD} + INCL	66.61/49 (0.05)	15.72/13 (0.26)	5.47/7 (0.60)	23.91/26 (0.58)	32.25/15 (0.01)
SF + F_A^{Deu} + INCL	81.91/49 (0.00)	23.94/13 (0.03)	10.01/7 (0.19)	35.08/26 (0.11)	38.83/15 (0.00)

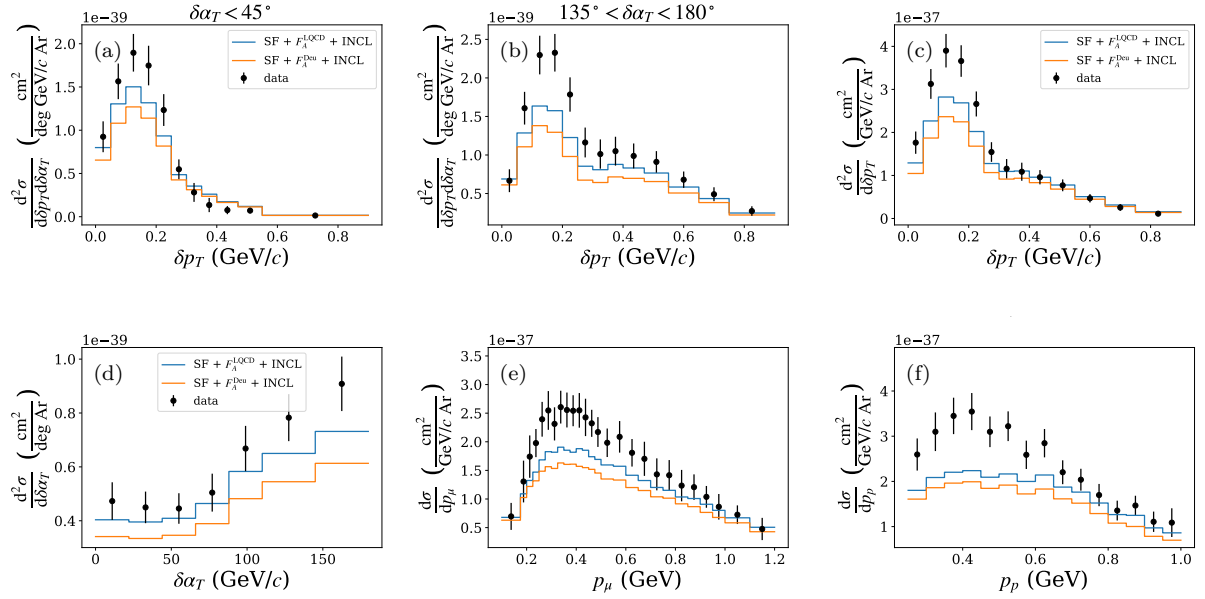


Figure 2: As in Fig. 1 but with the spectral-function ground state and INCL FSI held fixed: F_A^{LQCD} (blue) versus F_A^{Deu} (orange). The LQCD form factor again improves every observable (see Table 2), but with INCL FSI both predictions remain well below the data, so varying F_A alone does not reconcile the SF+INCL configuration with MicroBooNE.